MATH 111 Calculus 1

Logarithms

and logarithmic differentiation

Week 5 · Lecture 13 · §6.3, §6.4

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

STAND UP

"For every $x \neq 0$: $\ln\left(x^2\right) = 2\ln x$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE. At $x = -1$ the left side is $\ln 1 = 0$, but the right side, $2\ln(-1)$, is not defined. The law $\ln(x^r) = r\ln x$ is for $x > 0$. For every $x \neq 0$: $\ln(x^2) = 2\ln|x|$ — and the absolute value returns in frame 8.

Logarithms: the inverse of b^x

Let $b > 0$, $b \neq 1$. b^x is increasing ($b > 1$) or decreasing ($0 < b < 1$), so it is one-to-one, with range $(0, \infty)$. Its inverse function is the logarithm with base b, $\log_b$.

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EXAMPLES $\log_2 64 = 6$, since $2^6 = 64$; $\log_8 2 = \frac{1}{3}$, since $8^{1/3} = 2$; $\log_{10} 0.01 = -2$, since $10^{-2} = 0.01$; $\log_b 1 = 0$, since $b^0 = 1$.

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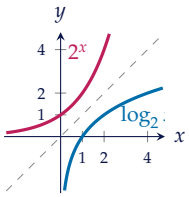
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Undoing $\log_b$ and b^x undo each other:

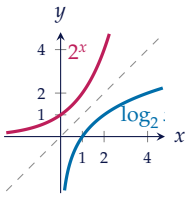
$\log_b(b^x) = x$ for every real x; $\quad b^{\log_b x} = x$ for every $x > 0$.

The graph of $\log_b$: the graph of b^x reflected in $y = x$. Domain $(0, \infty)$, range $\mathbb{R}$; continuous, being the inverse of a continuous function.



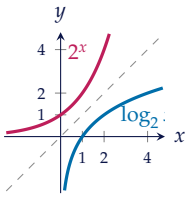
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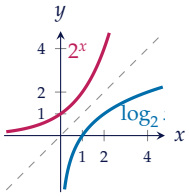


Laws of logarithms ($x, y > 0$, r real):

$$\log_b(xy) = \log_b x + \log_b y, \qquad \log_b \frac{x}{y} = \log_b x - \log_b y, \qquad \log_b(x^r) = r \log_b x.$$

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PROOF Put $u = \log_b x$, $v = \log_b y$; that is, $b^u = x$, $b^v = y$. Then $b^{u+v} = b^u b^v = xy$, $b^{u-v} = b^u / b^v = x/y$, $b^{ru} = (b^u)^r = x^r$. By the definition of $\log_b$: $\log_b(xy) = u + v$, $\log_b \frac{x}{y} = u - v$, $\log_b(x^r) = ru$. ∎

Caution. $\log_b(x + y)$ is not $\log_b x + \log_b y$: $\log_2(1 + 1) = 1$, while $\log_2 1 + \log_2 1 = 0$.

The natural logarithm

Definition $\ln x = \log_e x$: $\quad \ln x = y \iff e^y = x.$

So $\ln e = 1$, $\ln 1 = 0$, $e^{\ln x} = x$ for $x > 0$, $\ln\left(e^x\right) = x$ for every x.

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Every power through e: $b = e^{\ln b}$, so

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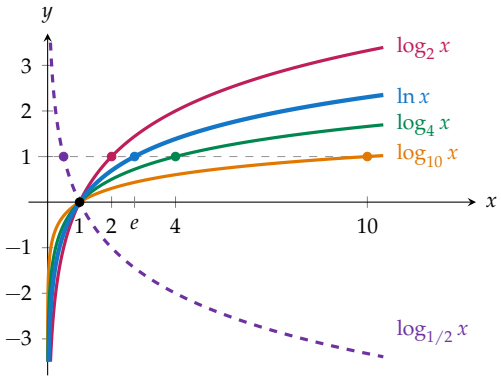
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Change of base $(b > 0, b \neq 1, x > 0)$: $\qquad \log_b x = \dfrac{\ln x}{\ln b}.$

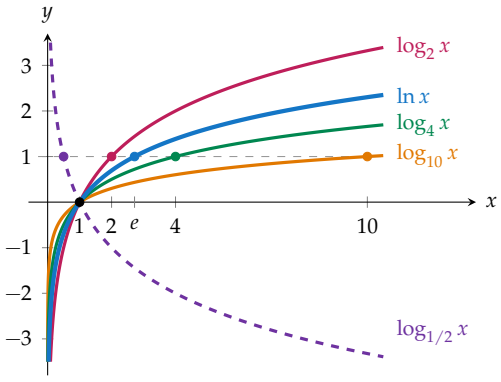
PROOF If $y = \log_b x$, then $b^y = x$. Take ln of both sides and use the third law: $y \ln b = \ln x$. Since $b \neq 1$, $\ln b \neq 0$, and so $y = \dfrac{\ln x}{\ln b}$. ■

Logarithms of different bases



All pass through $(1, 0)$: $\log_b 1 = 0$.

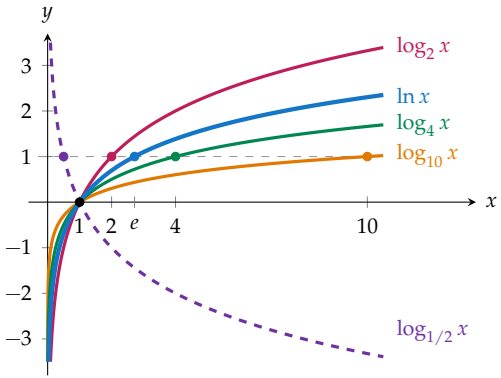
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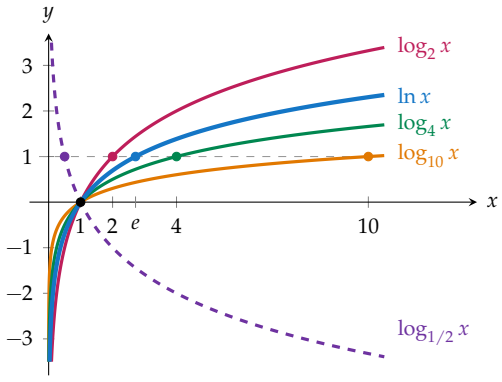
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$\log_{1/2} x = -\log_2 x$: the mirror image in the x-axis.

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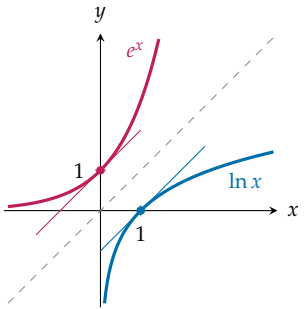
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Why: $\log_b x = \dfrac{\ln x}{\ln b}$. Every graph is the graph of $\ln x$ scaled vertically by the factor $\frac{1}{\ln b}$ — and flipped when $b < 1$.

The derivative of $\ln x$

Theorem $\ln$ is differentiable on $(0, \infty)$, and $\dfrac{d}{dx} \ln x = \dfrac{1}{x}$.



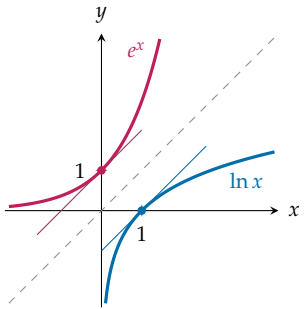
slope 1 at $(0, 1)$ on e^x; mirrored: slope 1 at $(1, 0)$ on $\ln x$

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PROOF $\ln$ is the inverse of $f(x) = e^x$, which is one-to-one and differentiable on $\mathbb{R}$, with $f'(x) = e^x \neq 0$. By the formula for the derivative of an inverse function, $(f^{-1})'(a) = \dfrac{1}{f'\big(f^{-1}(a)\big)}$, for every $a > 0$:

$$(\ln)'(a) = \frac{1}{f'(\ln a)} = \frac{1}{e^{\ln a}} = \frac{1}{a}.$$

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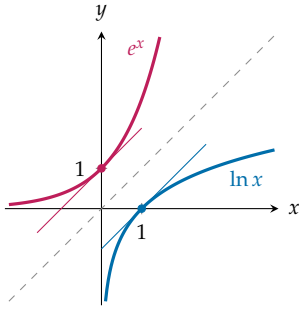
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Another proof, by the chain rule: $y = \ln x$ means $e^y = x$. Differentiate: $e^y y' = 1$, so $y' = \dfrac{1}{e^y} = \dfrac{1}{x}$.



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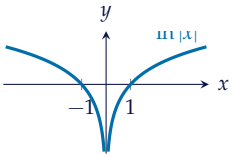
With the chain rule (g differentiable, $g(x) > 0$; $\ln'(u) = \frac{1}{u}$):

$$\frac{d}{dx}\ln\big(g(x)\big) = \underbrace{\ln'}_{\substack{\text{derivative}\\ \text{of outside}}} \underbrace{\big(g(x)\big)}_{\substack{\text{evaluated}\\ \text{at inside}}} \cdot \underbrace{g'(x)}_{\substack{\text{derivative}\\ \text{of inside}}} = \frac{1}{g(x)} \cdot g'(x) = \frac{g'(x)}{g(x)}.$$

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$\ln|x|$ is defined for $x \neq 0$; its derivative is $\frac{1}{x}$. **PROOF** For $x > 0$, $\ln|x| = \ln x$: derivative $\frac{1}{x}$. For $x < 0$, $\ln|x| = \ln(-x)$: derivative $\frac{1}{-x} \cdot (-1) = \frac{1}{x}$. ■

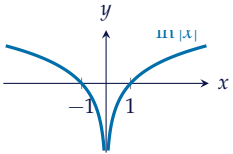


two branches; slope $\frac{1}{x}$ on both

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For every $x \neq 0$ $\big(\ln|x|\big)' = \frac{1}{x}$; and $\big(\ln|g(x)|\big)' = \frac{g'(x)}{g(x)}$ where $g(x) \neq 0$.

Three examples

(a) $y = \ln(3x^2 + 1)$: $\quad y' = \dfrac{1}{3x^2+1} \cdot 6x = \dfrac{6x}{3x^2+1}$.

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$$y' = \underbrace{3}_{\text{derivative of outside}} \Big(\underbrace{\ln x}_{\text{evaluated at inside}} \Big)^2 \cdot \underbrace{\tfrac{1}{x}}_{\text{derivative of inside}} = \frac{3(\ln x)^2}{x}.$$

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(c) $y = \ln \dfrac{x^2}{(x+3)^4}$, $x \neq 0, -3$. **Laws first:** $y = 2\ln|x| - 4\ln|x+3|$ (the warm-up's absolute values).

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(c) $y = \ln \dfrac{x^2}{(x+3)^4}$, $x \neq 0, -3$. **Laws first:** $y = 2\ln|x| - 4\ln|x+3|$ (the warm-up's absolute values).

Then $y' = \dfrac{2}{x} - \dfrac{4}{x+3} = \dfrac{6-2x}{x(x+3)}$. The quotient rule inside a chain would have taken a page.

Question 36

For $x > 0$: $\dfrac{d}{dx}\ln(5x) = \ ?$

A. $\dfrac{5}{x}$

B. $\dfrac{1}{5x}$

C. $\dfrac{1}{5} + \dfrac{1}{x}$

D. $\dfrac{1}{x}$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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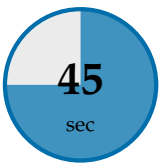
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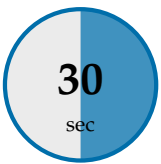
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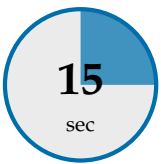
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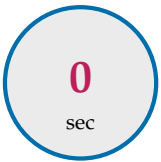
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

For $x > 0$: $\dfrac{d}{dx} \ln(5x) = \ ?$

A. $\dfrac{5}{x}$

B. $\dfrac{1}{5x}$

C. $\dfrac{1}{5} + \dfrac{1}{x}$

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

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C. $\dfrac{1}{5} + \dfrac{1}{x}$

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Hands down. Thank you.

A B C D

Question 36

EXPLAIN

For $x > 0$: $\dfrac{d}{dx}\ln(5x) = \ ?$

A. $\dfrac{5}{x}$

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

For $x > 0$: $\dfrac{d}{dx}\ln(5x) = \ ?$

A. $\dfrac{5}{x}$

B. $\dfrac{1}{5x}$

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D. $\dfrac{1}{x}$

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

D. Chain rule: $\dfrac{1}{5x} \cdot 5 = \dfrac{1}{x}$. Or laws first: $\ln(5x) = \ln 5 + \ln x$, and $\ln 5$ is a constant. B forgets the inside's derivative. A evaluates the outside's derivative, $\frac{1}{u}$, at x instead of at the inside $5x$: $\frac{1}{x} \cdot 5$. C differentiates the constant $\ln 5$ as if it were $\ln x$. The graph of $\ln(5x)$ is the graph of $\ln x$ moved up by $\ln 5$: the same slopes.

The derivatives of b^x and $\log_b x$

For $b > 0$: $\dfrac{d}{dx} b^x = b^x \ln b.$

PROOF $b^x = e^{x \ln b}$; by the chain rule,
$\left(e^{x \ln b}\right)' = e^{x \ln b} \cdot \ln b = b^x \ln b.$ ■

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So the slope of b^x at 0 has a name: $\lim_{h \to 0} \dfrac{b^h - 1}{h} = \ln b.$ $\ln 2 = 0.693147\ldots$, $\ln 3 = 1.098612\ldots$

Only one base has slope 1 at 0: the base e. **PROOF** If the slope of b^x at 0 is 1, then $\ln b = 1 = \ln e$, and ln is one-to-one, so $b = e$. ■

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For $b > 0$, $b \neq 1$, $x > 0$:
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PROOF $\log_b x = \dfrac{\ln x}{\ln b}$, and $\ln b$ is a constant. ■

Why calculus prefers e: $\ln e = 1$, and the extra factor disappears from both formulas.

Question 37

$$\frac{d}{dx}\left(x^3 + 3^x + 3^3\right) = \ ?$$

A. $3x^2 + x \cdot 3^{x-1}$

B. $3x^2 + 3^x$

C. $3x^2 + 3^x \ln 3$

D. $3x^2 + 3^x \ln 3 + 27$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

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60 sec

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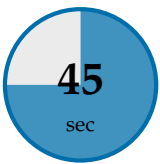
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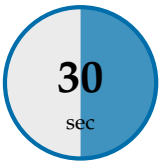
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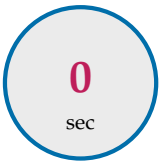
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Answer

C. x^3: the base varies, the exponent is fixed — the power rule, $3x^2$. 3^x: the base is fixed, the exponent varies — $3^x \ln 3$. $3^3 = 27$ is a constant: derivative 0. A uses the power rule on 3^x, as if the exponent were fixed. B forgets $\ln 3$ — right only for base e. D differentiates the constant 3^3 as if it were x^3.

Logarithmic differentiation

Three steps **1.** Take ln of both sides of $y = f(x)$; expand by the laws.
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Without logarithms: the quotient rule around a product, with a chain in each factor.
Where y can be negative, take $\ln|y|$: the same formulas (frame 8).

Variable base, variable exponent

base, exponent	example	derivative
constant, constant	b^n	0
variable, constant	$\big[f(x)\big]^n$	$n\big[f(x)\big]^{n-1} f'(x)$ (power rule, chain)
constant, variable	$b^{g(x)}$	$b^{g(x)} \ln b \cdot g'(x)$ (frame 13, chain)
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The right answer is the sum of two wrong answers: exponent frozen (power rule), $x(x^2+1)^{x-1}\cdot 2x$; base frozen (frame 13), $(x^2+1)^x\ln(x^2+1)$.

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For $x < 0$ and $n = \frac{p}{q}$ with q odd (as for $n = \frac{1}{3}$): logarithmic differentiation with $\ln|y|$. At $x = 0$: the definition.

The number e as a limit

Theorem
$e = \lim_{x \to 0} (1+x)^{1/x}$.

PROOF The slope of ln at 1 is $\frac{1}{1} = 1$; from the definition of the derivative:

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(rounded to 6 decimals)

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With $x = \frac{1}{n}$: $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$ (given ε, the δ of the limit above serves every $n > \frac{1}{\delta}$) — the compound interest of Bernoulli's paper of 1690.

A little history · 1614–1748

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1615. **Henry Briggs** visits Napier in Edinburgh; they agree that the logarithm of 1 should be 0, and on base 10. Briggs's *Arithmetica Logarithmica* (1624): the logarithms of 1 to 20 000 and of 90 001 to 100 000, to 14 decimal places. **Jost Bürgi** had made tables of his own; they were printed in Prague in 1620.

A little history · 1614–1748

1614. John Napier publishes the *Mirifici Logarithmorum Canonis Descriptio*: tables that turn multiplication into addition. The word is his — *logos*, proportion, and *arithmos*, number. His logarithms are "not really to any base", and his logarithm of 1 is not 0.

1615. Henry Briggs visits Napier in Edinburgh; they agree that the logarithm of 1 should be 0, and on base 10. Briggs's *Arithmetica Logarithmica* (1624): the logarithms of 1 to 20 000 and of 90 001 to 100 000, to 14 decimal places. **Jost Bürgi** had made tables of his own; they were printed in Prague in 1620.

1649. de Sarasa relates logarithms to areas under the hyperbola $xy = 1$. **Euler** (1748) writes that the logarithms to base e are called "natural or hyperbolic" — the names were already in use. Why areas under $1/x$ are logarithms: $(\ln x)' = 1/x$, and the Fundamental Theorem of Calculus.

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This week:

Homework: §6.3 and §6.4 — read and work out the examples. Next: inverse trigonometric functions and hyperbolic functions (§6.6, §6.7).

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: "$\frac{d}{dx}\ln(5x)$ *is* $\frac{1}{x}$*, and not* $\frac{5}{x}$*, because . . . "*
   (one sentence, or one line of algebra).
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.